COHERENCE · EVALUATION

App Evaluation Report

Your AI path to instant clarity — what works, what was tested, what is still open

22 October 2026 · https://your-ai-learning-partner.lovable.app

SUMMARY

Coherence is a timestamp-aware AI learning companion. A learner watches (or reads) a lesson, asks an AI tutor questions answered only from the parts of the transcript they have actually covered, takes quizzes per topic, revises missed concepts on a spaced-repetition schedule, and finishes with a three-question mock interview that produces an evidence-backed scorecard shareable through a read-only link.

The app works on phones (bottom navigation) and on computers (left sidebar, wider layouts). All core flows were exercised end to end in a browser; the automated suite passes 10/10.

10 / 10	6 / 6	19	42
automated tests passing	security checks denied	quiz questions	transcript chunks

1. Feature checklist

Area	Feature	Status
Accounts	Email/password, Google, and Sign in with Lovable	**Working**
Workspace	YouTube player with saved watch progress and topic markers	**Working**
Workspace	Timestamp jumps (queued until the player is ready)	**Working**
Tutor	AI chat grounded in watched transcript segments only	**Working**
Tutor	Citations verified server-side against watched chunks	**Working**
Quiz	Per-topic multiple-choice questions, server-validated answers	**Working**
Revision	Spaced-repetition schedule built from wrong answers	**Working**
Interview	3-question mock interview, 4 scores (1–5) with verbatim evidence	**Working**
Interview	Shareable read-only scorecard link	**Working**
Lessons	Import your own lesson (SRT/VTT/text transcript)	**Working**
Lessons	Admin-only replacement of the sample transcript	**Working**
Reading	Ordered “mark segment read” progress for transcript-only lessons	**Working**
History	Practice, revision, and interview history	**Working**
Voice	Microphone input via Web Speech API	**Built; not yet verified with a real microphone**

2. Automated test results

Test suite run on 22 October 2026 with **vitest**:

Test file	Tests	Result
youtube-player.test.tsx	4	**Passed**
learning-rules.test.ts	5	**Passed**
app-routing.test.tsx	1	**Passed**
Total	**10**	**10 passed, 0 failed**

Covered rules include: quiz answers are validated on the server, revision intervals follow the spaced-repetition schedule, tutor citations must match watched transcript chunks exactly, and the video player queues timestamp requests until it is ready.

3. Manual end-to-end checks (browser)

- Sign-in, lesson loading, watch progress saving, and timestamp jumps — passed.
- Quiz submission saves attempts and creates revision items for wrong answers — passed.
- Mock interview start, resume from a saved session, completion, and scorecard generation — passed.
- Public scorecard share link opens read-only without sign-in — passed.
- Lesson import with a user-supplied transcript — passed.
- Phone layout: no horizontal overflow; bottom navigation usable — passed.

4. Security and data-protection checks

Check	Result
Non-admin user tries to replace the sample transcript	**Denied**
Client tries to read quiz answer keys directly	**Denied — question-only columns exposed**
Client tries to write a forged “completed” interview score	**Denied — server-only privileged write after ownership check**
Seeking the video ahead to unlock unseen transcript content	**Denied — server-bounded watch progress**
Skipping transcript segments in reading mode	**Denied — ordered reading RPC**
Shared scorecard link	**Read-only; no private activity exposed**

Row-level security is enabled on every table, and all AI calls run through authenticated server functions with rate limiting.

5. Current usage data (from the database, 22 Oct 2026)

- Lessons: 2 (1 seeded MIT physics lesson, 1 imported test lesson)
- Topic markers: 9 · Transcript chunks: 42 · Quiz questions: 19
- Quiz attempts: 3 (1 correct) · Revision items: 2
- Interview sessions: 4 · Progress records: 3

6. Known limitations

1. **Sample transcript is approximate.** The seeded MIT lecture transcript was written for demonstration and is not the official caption file. An authentic transcript can be supplied through the admin “Replace sample transcript” action.
2. **Microphone input untested.** Voice answering uses the browser's Web Speech API; it has not been verified with a real microphone.
3. **AI provider state.** Interview and tutor generation depends on the AI gateway; an earlier access/credit issue was observed and resolved during testing.

7. Conclusion

All core learning flows — watch, ask, quiz, revise, interview, share — work end to end on phone and computer layouts, the automated suite passes 10/10, and the security checks confirm that answers, scores, and private activity cannot be forged or leaked from the browser. The two remaining items are content-related (authentic captions) and hardware-related (microphone verification), not code defects.

Accompanying dataset: coherence_sample_dataset.pdf · coherence_sample_dataset.zip